



A physics-based algorithm to perform predictions in football leagues

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ABSTRACT

In this work, we extended a stochastic model for football leagues based on the team's potential (da Silva et al., 2013) for making predictions instead of only performing a successful characterization of the statistics on the punctuation of the real leagues. Our adaptation considers the advantage of playing at home when considering the potential of the home and away teams. The algorithm predicts the tournament's outcome by using the market value or/and the ongoing team's performance as initial conditions in the context of Monte Carlo simulations. We present and compare our results to the worldwide known SPI predictions performed by the "FiveThirtyEight" project. The results show that the algorithm can deliver good predictions even with a few ingredients and in more complicated seasons like the 2020 editions where the matches were played without fans in the stadiums.

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1. Introduction

Football is the most famous sport globally and is professionally performed in more than 200 countries according to FIFA¹ generating many jobs around the world, both directly and indirectly. Despite this sport seems frivolous to a part of the world's population and socially unequal when we look at the amount of money some superstar players receive in major championships [1,2] in comparison with players from many teams of country-side cities in Latin America as well as in other undeveloped countries, as a product of that same unequal society [3] it also generates many direct and indirect employments.

Called soccer in the USA, this sport, which is the world's most popular one, has other economic and social importance.² Translating these points into money is difficult, but we have a rough idea of how they can impact people's lives, especially the poorest ones since it can be seen as a great career opportunity for young citizens that grow up in underprivileged conditions such as in *favelas* in Brazil [4]. Not necessarily, for this reason, physicists have devoted some work and time to describe statistics related to football (see, for example, Refs. [5–7]) in order to understand the stochastic and deterministic aspects of this exciting research area.

Until the 1980s, the football's scoring system was 2-1-0, i.e., the teams received two points for a win, one point for a draw, and zero points for a loss. This system, however, had benefited many "cross-country" teams which played to draw

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¹ Fédération Internationale de Football Association — <https://www.fifa.com/>.

² An interesting case can be observed in this technical report about the Spanish league <https://assets.kpmg/content/dam/kpmg/pdf/2015/07/socioeconomic-impact-professional-football.pdf>.

since this was very advantageous in that scoring system. However in Ref. [8] the authors show that system 2-1-0 is more efficient than 3-1-0 to predict the “best” team.

To make the game more attractive and, in a way, more competitive, an evolution has started since then. The scoring system 3-1-0 prevailed over the previous one, as well as other improvements in the rules of the game. For example, when the goalkeeper receives the ball from one of its teammates, he/she is prohibited from catching it with his/her hands. This simple rule increased the time of the ball in motion during matches. Such rules are updated from time to time and all improvements (or throwbacks), as well as new rules, can always be found in the webpage related to “laws of the game” in the international football association board (IFAB).³

The magic point of football, which also occurs in many other sport modalities, is that the best team does not always win the match, and in some championships, “dark horses”, a term used to describe a little known candidate or competitor, who unexpectedly wins or succeeds, are ubiquitous. For instance, in Refs. [9,10], the authors showed that league-type tournaments are inefficient and less fair in comparison with elimination-type tournaments due to these upsetting victory effects.

In Brazil, this “upset victory” is called a “zebra”, a name created by Gentil Cardoso, a Portuguese iconic football coaching who told a reporter during an interview about the chances of his team beating Vasco da Gama in a match valid for the regional championship in Rio de Janeiro, Brazil. Cardoso, at that time, used the word “zebra” to refer to an animal not included in a popular (but prohibited) gambling in Brazil, which can be freely translated as “animal game” in a free translation.

The so-called “zebras” are more frequent than one could imagine [11,12] and many times, they break the deterministic characteristics of championships, making the favorite teams not always the champions.⁴

Commonly, the champion team has a good cast of players, which in turn, depends, for instance, on the financial strength [13] of the club (although not always). In balanced championships like those in Brazil, where many teams have already become champions, the symbiosis of the team’s players and a good coach can even change history.

The ability to predict the result of a match between two teams as well as the league champion has been widely studied recently. Many factors can contribute to a short-term forecast, such as injuries, tactics, players’ skills, etc. Statistical [14] and machine learning [15,16] frameworks have been proposed to make sports prediction as well.

Based on these assumptions, in this paper, we intend to answer the following question: Are we able to predict champions and relegations in a football championship using simple mathematical models?

To shed light on this question, we first consider that each team can beat its opponent, which, in turn, is translated as the probability of victory. This probability depends on amounts defined as the team’s potentials. These potentials consider several factors, for example, the economic power of the team and its history of successes along the championship. This study also takes into consideration a previous model that successfully described the statistics of team scores in different double round-robin system (DRRS) championships [12,17,18], i.e., the championships in which all teams play to each other twice, in turn, and return. In addition, we will study different possible evolutions in order to predict, for instance, who was champion the most times in the simulated championships.

The team’s potential is incremented/decremented according to the match result. Higher potentials mean higher win probabilities in a kind of “preferential attachment”, i.e., the known scheme proposed by Barabási and Albert [19] to grow graphs that follow power-law distribution for the nodes, which increments the degree of the nodes. However, in the case of football, one can also decrement the team’s potential (nodes) differently from what occurs in the preferential attachment.

One of the authors of this paper, in a recent publication [20], explored a kind of random walk which tends to resort to more visited nodes. In that case, the potential of a node corresponded to its attractiveness which are always incremented when the node is visited. This is also an allusion to the work of Barabási and Albert, which showed that such mechanisms can involve many contexts in the physics of complex systems.

On the other hand, if average values substitute the team’s potential, these scoring schemes based on probabilities that depend on the team’s potential also lead us to imagine that teams’ scores can work as a diffusive process. If the team’s potentials are constant values during the evolution, and by consequence, the draw probability r is also constant, the probability $P(x, t)$ of a team having score x at time t is the solution of the partial differential equation [17]:

$$\frac{\partial P}{\partial t} = - \left[r + 3 \frac{(1-r)}{2} \right] \frac{\partial P}{\partial x} + 3 \frac{(1-r)}{2} \frac{\partial^2 P}{\partial x^2} - \frac{(1-r)}{2} \frac{\partial^3 P}{\partial x^3} \quad (1)$$

which results in a super-diffusive behavior since it is verified that:

$$\langle x^2 \rangle = \int_0^\infty P(x, t) x^2 dx \sim t^\Lambda$$

³ <https://www.theifab.com/history/ifab>.

⁴ Ref. [11] is related to a report about football organized by I. Zolkernevic, highlighting the contribution in this area of some other Brazilian authors. In addition to us, it includes comments about the work of Ribeiro et al. (see also [7]). The direct link to this report is <https://revistapesquisa.fapesp.br/en/upset-victories-common-brazilian-soccer-championship>.

with $\Lambda > 1$ according to the numerical solution of Eq. (1), which corroborates with the results from real championships [17]. By considering the change of the team's potential over time, we expect an even richer behavior of the system during the evolution.

This paper proposes a model that makes predictions of the champion team, the four best teams (G4), and the four worst teams (Z4) in a championship. For this purpose, we adopted two different parameters: the market value and the teams' performance, as shown in Section 2. In Section 3, we divide our results into two different parts: the first one corresponds to an exploration of the tuning/calibration of some parameters with particular attention to the influence of market value and memory effects on modeling and, in the second part of the results, we take into account the optimal parameters obtained previously to test our model presenting the predictions for the Brazilian, Italian, German, Spanish, and English football leagues. We performed a detailed comparison with some specialized websites. Finally, some conclusions are presented in Section 4.

2. Model and optimization

Our model is based on the agent-based model [12,17,18] that considers a system of N teams playing against each other according to DRRS.⁵ Every agent represents a team with a pair of potentials ϕ and ψ (as host and visitor, respectively). Each agent interacts with only one different agent at each time step (round) in a predefined sequence (scheduled matches for the season).

After a given season starts and $k - 1$ rounds have been played, one supposes that the team i plays as host (*home* team) against the team j as visitor (*away* team) at the k th round. Based on the fact that the number of goals in a match follows a Poisson distribution [6,21,22] (one shows that other scored-based sports such as basketball [23], and ice hockey [24] also present similar scoring dynamics), the probability of the match resulting in a draw as a conditional probability function [12] of potentials of the teams involved is given by:

$$\begin{aligned} r_{draw}^{(i,j)}(k) &= \Pr \left[(g_i = g_j) | (\phi_k^{(i)}, \psi_k^{(j)}) \right] \\ &= \sum_{g=0}^{\infty} \frac{(\phi_k^{(i)} \psi_k^{(j)})^g}{g!^2} e^{-(\phi_k^{(i)} + \psi_k^{(j)})} \\ &= e^{-(\phi_k^{(i)} + \psi_k^{(j)})} I_0 \left(2\sqrt{\phi_k^{(i)} \psi_k^{(j)}} \right) \end{aligned} \quad (2)$$

where g is the number of goals scored by each team, ϕ and ψ correspond to the potentials of the host and the visitor teams, respectively, and

$$I_\nu(z) = \left(\frac{z}{2}\right)^\nu \sum_{n=0}^{\infty} \frac{(\frac{1}{4}z^2)^n}{n! \Gamma(\nu + n + 1)}$$

is the modified Bessel function of the first kind. Our reasons for defining ϕ and ψ , is due to the historical statistical advantage of host teams over the visitors [25–27]. That happens for a number of reasons such as traveling distances [28], crowd induced referee bias [29,30], altitude of the home stadium [31], etc. Playing at home can be seen as an environmental advantage capable of leveling teams with different technical skills/investments such as shown in Ref [32]. The probability that a team i has of winning the match against a team j at q th round considers the product of the complement of Eq. (2) and a factor that weights the potentials of the teams involved such as follows:

$$\Pr(g_i > g_j, q) = \left[1 - r_{draw}^{(i,j)}(q) \right] \cdot \frac{\phi_q^{(i)}}{\phi_q^{(i)} + \psi_q^{(j)}}. \quad (3)$$

The chances of the team j winning the match is defined by the complement:

$$\begin{aligned} \Pr(g_i < g_j, q) &= 1 - \Pr(g_i > g_j, q) - r_{draw}^{(i,j)}(q) \\ &= \left[1 - r_{draw}^{(i,j)}(q) \right] \cdot \frac{\psi_q^{(j)}}{\phi_q^{(i)} + \psi_q^{(j)}}. \end{aligned} \quad (4)$$

The team's potential is updated after each match. Therefore, we must consider the final result of the match, i.e., whether the team won, lost or if the match ended in a draw. With this in mind, we defined that the winner and the loser teams

⁵ https://en.wikipedia.org/wiki/Round-robin_tournament.

will have their potentials changed respectively by the quantities $\Delta\phi_q$ and $\Delta\psi_q$. By considering that the home team i plays against the visiting team j , we have

$$\Delta\phi_q^{(i)} = \begin{cases} -\Delta\psi_q^{(j)} = 3 & \text{if } g_i^{(q)} > g_j^{(q)}, \\ \Delta\psi_q^{(j)} = 1 & \text{if } g_i^{(q)} = g_j^{(q)}, \\ -\Delta\psi_q^{(j)} = -3 & \text{if } g_i^{(q)} < g_j^{(q)}, \end{cases} \quad (5)$$

yielding $\Delta\psi_q^{(i)} = \Delta\phi_q^{(j)} = 0$ in all conditions.

Before any season starts, the market value¹ of each team is supposed to be one of the main factors to reflect its future performance, which has influence on its chances of becoming the future champion of the league. After the start of a season, the real performance of the teams comes to light as new and important information that must be taken into account in predictive algorithms. Thus, we define both potentials of a team i at the q th round as:

$$\begin{aligned} \phi_q^{(i)} &= \phi_{k_0}^{(i)} + \gamma \sum_{j=k_0}^{q-1} \Delta\phi_j^{(i)}, \\ \psi_q^{(i)} &= \psi_{k_0}^{(i)} + \gamma \sum_{j=k_0}^{q-1} \Delta\psi_j^{(i)}, \end{aligned} \quad (6)$$

where $\phi_{k_0}^{(i)} \equiv \alpha M^{(i)} + \beta P_{k_0|\phi}^{(i)}$ and $\psi_{k_0}^{(i)} \equiv \alpha M^{(i)} + \beta P_{k_0|\psi}^{(i)}$ are the initial potentials, $M^{(i)}$ is the market value of team i , and therefore, α is the market coefficient.

In this work, we used the actual scored points of teams as input data to describe real world performance. However, our model has flexibility in the sense that we can use any other observable performance parameter such as number of goals [33], chances of goals [34], etc. Our model also allows to use ϕ_{k_0} and ψ_{k_0} as function of a team's single performance index as proposed in [35]. However, we choose to define ϕ_{k_0} and ψ_{k_0} as a linear combination of an objective performance parameter (scored points for simplicity) and a subjective performance parameter (market value). The performance coefficient β is associated to the scored points and the quantities $P_{k_0|\phi}^{(i)}$ and $P_{k_0|\psi}^{(i)}$ are the cumulative scored points of team i playing as host and its cumulative scored points when it plays as visitor, respectively. Obviously, $P_{k_0}^{(i)} = P_{k_0|\phi}^{(i)} + P_{k_0|\psi}^{(i)}$ is the total scored points.

Finally, we complete the description of the potentials by addressing the last term of Eq. (6), which is the sum of the successive potential increments/decrements from the round k_0 up to the q th round, i.e., $\sum_{j=k_0}^{q-1} \Delta\phi_j^{(i)}$ and $\sum_{j=k_0}^{q-1} \Delta\psi_j^{(i)}$. Thus, γ is a coefficient associated to the evolution of the team's potential.

Now, it comes the problem: what are the values of α , β , and γ that best describe a real football championship according to our model? To reach this goal, if $k-1$ rounds have been played on the ongoing season, we have to test different values of the potentials' coefficients over $\Delta k = k - k_0$ rounds by using the known outcomes of played matches. Thus, "memorizing" the time interval works as a "learning process" used to determine the values $\alpha = \alpha_{opt}$, $\beta = \beta_{opt}$, and $\gamma = \gamma_{opt}$, which maximizes some relevant parameters by comparing results from the simulation and real-world ongoing season.

Once we know the "optimal" coefficients, we can simulate our model from the current unplayed round to the end of the season to get our predictions: the winner of the championship, as well as the four best teams (G4) and the four worst teams (Z4) of the championship. The flowchart in Fig. 1 summarizes the whole process.

We technically detail the optimization procedure and the final simulations in the following algorithms. The main algorithm (Algorithm 1) defines the evolution of the potentials described by Eq. (6) in a general way, as well as the evolution of the scores.

First, it is necessary to perform the optimization. We must decide which measurable quantity we want to find its optimal value. The fact that we are studying recent historical rounds (previously to the current one) that we already know their outcomes let us define the *matching hits* metrics denoted by:

$$\xi_q = \frac{2}{N} \sum_{l=1}^{N/2} \delta(\varphi_q^{(l)}|_S, \varphi_q^{(l)}|_R) \quad (7)$$

where $\varphi_q^{(l)}|_S$ and $\varphi_q^{(l)}|_R$ are the results of the l th match in the q th round of the simulation (S) and the real world (R), respectively, and $\delta(x, y)$ is the well-known Kronecker symbol. Here, we are considering only the results of victory, defeat, or draw, without considering the exact number of goals. Thus, in the tuning algorithm (Algorithm 2), presented below, we perform the optimization process to obtain α_{opt} , β_{opt} , and γ_{opt} , i.e., the values of α , β , and γ that maximizes ξ_q .

¹ <https://www.transfermarkt.com/>.

Ongoing season currently at round k

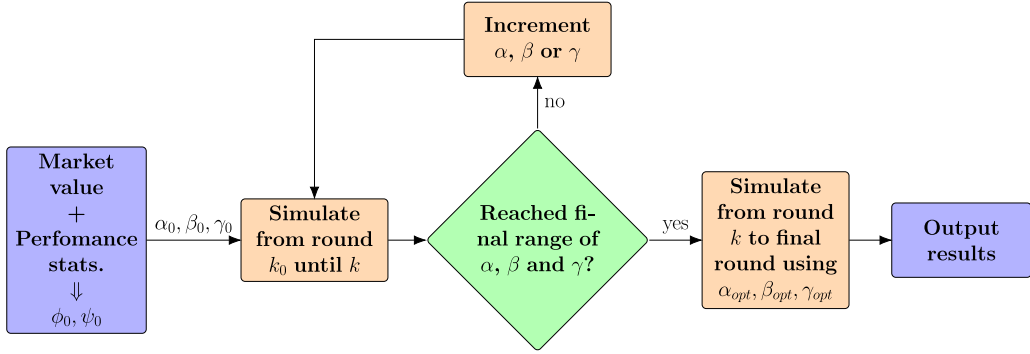


Fig. 1. Flowchart detailing the algorithm used to make predictions based on our model. From left to right, we first add real data input to the model. Then we search for the coefficients that best fit the model and finally, we simulate the championship and get the prediction results.

Algorithm 1: Main Algorithm

```

1: if (rand[0, 1] < rdraw) then
2:   P(i) = P(i) + 1, P(j) = P(j) + 1; φ(i) = φ(i) + γ * Δφ(i); ψ(j) = ψ(j) + γ * Δψ(j);
3: else
4:   if (rand[0, 1] < (φ(i) / (φ(i) + ψ(j)))) then
5:     P(i) = P(i) + 3; φ(i) = φ(i) + γ * Δφ(i); ψ(j) = ψ(j) + γ * Δψ(j);
6:   else
7:     P(j) = P(j) + 3; φ(i) = φ(i) + γ * Δφ(i); ψ(j) = ψ(j) + γ * Δψ(j);
8:   end if
9: end if
  
```

Algorithm 2: Tuning

```

1: for α = 0 to αmax step Δα do
2:   for β = 0 to βmax step Δβ do
3:     for γ = 0 to γmax step Δγ do
4:       for irun = 1 to Nrun do
5:         for iround = k0 to k do
6:           run Main Algorithm;
7:         end for
8:       end for
9:       if ξ > ξmax then
10:        αopt = α; βopt = β; γopt = γ; ξmax = ξ;
11:       end if
12:     end for
13:   end for
14: end for
  
```

The latest algorithm (Algorithm 3), shown in the following, performs the final simulations that will result in the predictions themselves. In this part of the algorithm, we perform N_{run} simulations using the optimal coefficients obtained in the previous part of the process (Algorithm 2). Each simulation results in a league table (standings) where teams are ranked according to their final points. The computing of the probabilities of each team becoming champion, qualifying

to the continental tournament, or relegating to a lower division consists of taking the number of simulations for each outcome and dividing them by the total number of simulations.

Algorithm 3: Final evolution

```

1: for  $i_{run} = 1$  to  $N_{run}$  do
2:   for  $i_{round} = k$  to  $N_{last}$  do
3:     run Main Algorithm using  $\alpha_{opt}$ ,  $\beta_{opt}$ , and  $\gamma_{opt}$ ;
4:   end for
5: end for

```

The purpose of each of these algorithms can be summarized as follows:

1. Algorithm 2 calls Algorithm 1 N_{run} times for each (α, β, γ) in the chosen range. Then, simulations are performed from round k_0 up to the k th round using $\phi_{k_0}^{(i)}$ and $\psi_{k_0}^{(i)}$ as input for $i = 1, \dots, N$ in order to obtain the average value of ξ_q for $\frac{N}{2} \Delta k$ matches.
2. By performing three external loops, Algorithm 2 increments the values of α , β , and γ by the amount $\Delta\alpha$, $\Delta\beta$, and $\Delta\gamma$, and then we repeat the step 1 for each set α , β , and γ in the chosen range until we find $(\alpha, \beta, \gamma) = (\alpha_{opt}, \beta_{opt}, \gamma_{opt})$.
3. With α_{opt} , β_{opt} , and γ_{opt} in hand, we perform N_{run} simulations once again, but now from round k to the last round ($N_{last} = 2(N - 1)$, where N is the number of teams) to obtain the average results of each team (how many times each team has become the champion, stood in g_4 zone, or stood in the z_4 zone), according to Algorithm 3 (final evolution).

In addition, we can use an alternative metric for the optimization process, which consists of considering two different quantities: the scored points, μ , and the position in the current standings (ranking), ν . The first quantity measures the difference between a team's scored points in the simulation (S) and those obtained in real-world (R):

$$\mu_q = \frac{1}{N} \sum_{l=1}^N \eta(P_q^{(l)}|_S, P_q^{(l)}|_R), \quad (8)$$

where $P_q^{(l)}|_S$ and $P_q^{(l)}|_R$ denote the total scored points in the simulation and real-world, respectively, for the team l at the q th round. We also define

$$\eta(x, y) = \begin{cases} 1 & \text{if } |x - y| \leq \epsilon, \\ 0 & \text{otherwise,} \end{cases}$$

where ϵ is an error parameter. If $\epsilon = 0$, we have what we called the *strong* predictor. On the other hand, $\epsilon = 1$ is called *medium* predictor, and $\epsilon = 2$ means *weak* predictor. The ranking-based variable takes into account the difference between the team's ranking position in the simulated current standings and the position in the real league at that point as follows

$$\nu_q = \frac{1}{N} \sum_{l=1}^N \eta(Q_q^{(l)}|_S, Q_q^{(l)}|_R), \quad (9)$$

where $Q_q^{(l)}$ denotes the ranking position of the team l at the q th round.

Thus the optimization process consists of searching for the values of α , β , and γ that maximizes

$$\xi_q = (\mu_q + \nu_q) / 2, \quad (10)$$

which represents what we denote as *score-ranking metric*. This paper used $\epsilon = 0$, (strongest criterion) when we applied the score-ranking metric. For simplicity, we searched for optimal coefficients in the range $(0, 1]$ in the optimization process (Algorithm 2).

3. Results

We divided our results into two parts. In the first part, we show the results, exploring all details of our method for the Brazilian league. Since we present the details for this particular league, in the second part of our results, we offer the summarized results for the European leagues to complete the validation of our results

3.1. Brazilian league

Our first results show how the parameters and quantities behave in the optimization part of the process (Algorithm 2). For that, we have used our model to simulate past seasons of the Brazilian football league, also known as *Brasileirão*. In this

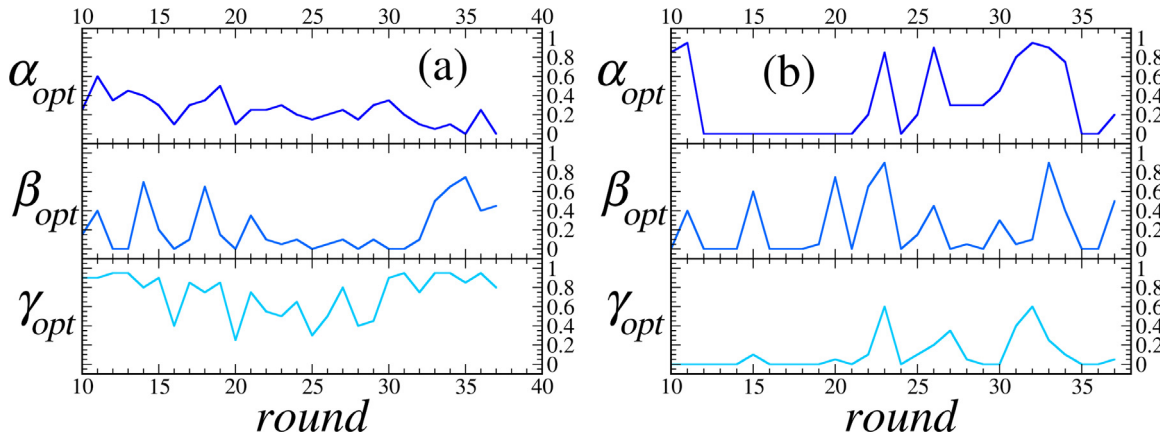


Fig. 2. Evolution of the parameters α_{opt} , β_{opt} , and γ_{opt} per round for the 2020 edition of the main Brazilian football championship. (a) Corresponds to the case where we used the matching hits metric and (b) corresponds to the score-ranking metric.

work, we obtained our results using $N_{run} = 100$ samples for both the optimization part (Algorithm 2) and the predictive simulations (Algorithm 3) whenever the values of the three coefficients are defined in order to obtain the optimal values. On the other hand, we always used $N_{run} = 1000$ samples when two coefficients, namely β and γ , are kept fixed. We also used $\Delta\alpha = \Delta\beta = \Delta\gamma = 0.05$ and $\Delta k = 10$ for all results unless stated otherwise.

Fig. 2(a) presents the evolution of the optimal coefficients when considering the matching hits metric and Fig. 2(b) takes into consideration the score-ranking metric for the 2020 edition of the *Brasileirão*. In the former figure, we observe that the coefficients seem to fluctuate slightly less than those obtained by the score-ranking metric (Fig. 2(b)), mainly when one observes the evolution of α_{opt} which reflects a decrease in dependence on market value as rounds are played.

An interesting finding is obtained when the values of β and γ are fixed (in this work they are set to 1) and we focus only on the evolution of α_{opt} in order to study how the market value contributes to our model's ability to make good predictions. This simplification allows us to implement our algorithm for $N_{run} = 1000$ samples without gaining much computational time. Fig. 3 shows the behavior of the quantity ξ_k for the matching hits metric (Eq. (7)) as function of α for different rounds of the 2020 (a), 2019 (b), 2018 (c), and 2017 (d) seasons of the *Brasileirão*. Particularly, for the season 2020, shown in Fig. 3(a), an “almost” monotonic decay is observed only for later rounds in the tournament, such as $k = 35$. For the other seasons, for at least $k \leq 30$, we are able to observe an initial increase and a subsequent decrease of ξ_k as a function of α , showing that ξ_k has a well defined local maximum value in the range $\alpha \in (0, 1]$. However, for the other seasons, we observe that this “almost” monotonic decay behavior appears sooner in the tournament, such as for round $k \geq 20$, indicating that the market value loses its influence over the matching hits metric as the championship evolves, corroborating the results shown in Fig. 2.

From now on, we focus our attention on the score-ranking metric (Eq. (10)). In Figs. 4(a) and 4(b), we show, respectively, how α influences the score and rank quantities (defined in Eqs. (8) and (9), respectively) for the 2020 *Brasileirão* season.

Differently from the matching hits metric, the variables used to compose the score-ranking metric present slow increasing tendency on α as can be observed in Fig. 4(a) for μ and (b) ν . Without loss of generality, in Fig. 4, we only presented the results for the 2020 season since it is the most debatable recent season due to the pandemic and its champion was incredibly decided only at the last round. Although not presented here, the other three previous seasons have similar behavior.

Once we show the way to obtain α_{opt} , it is interesting to present a typical plot of α_{opt} as function of round (k). In Fig. 5(a) and (b), we show our results when using the matching hits metric and the score-ranking metric, respectively, as criteria for the optimization part of the algorithm.

We can observe that α_{opt} has a clear tendency to decrease as k increases when using the matching hits criterion corroborating the results shown in Figs. 2 and 3. On the other hand, α_{opt} varies enormously with the score-ranking criterion, not showing any apparent connection with the importance of the market value on the predictions.

To complete the study of the optimization part of our model, we focus on the time interval (Δk) used to find the coefficients that best fit our model. One question we could raise here is: how much in the past (previous rounds) should we look so that our model can make sound predictions? To answer this question, we consider only the matching hits metric, which was the metric that showed a more reliable behavior for optimization purposes. Fig. 6 presents the maximum ξ_k , i.e., the value calculated at $\xi_{opt} = \xi(\alpha_{opt})$ as function of Δk for three different seasons of the *Brasileirão*: 2018, 2019, and 2020 at different rounds and keeping fixed β and γ to the unit.

This figure shows that ξ_{opt} presents small fluctuations around a mean value for the interval $6 < \Delta k < 20$, where one shows the results for the rounds from $k = 5$ to $k = 35$. For this reason, we have chosen $\Delta k = 10$ in the first part of the results and used it for the following results.

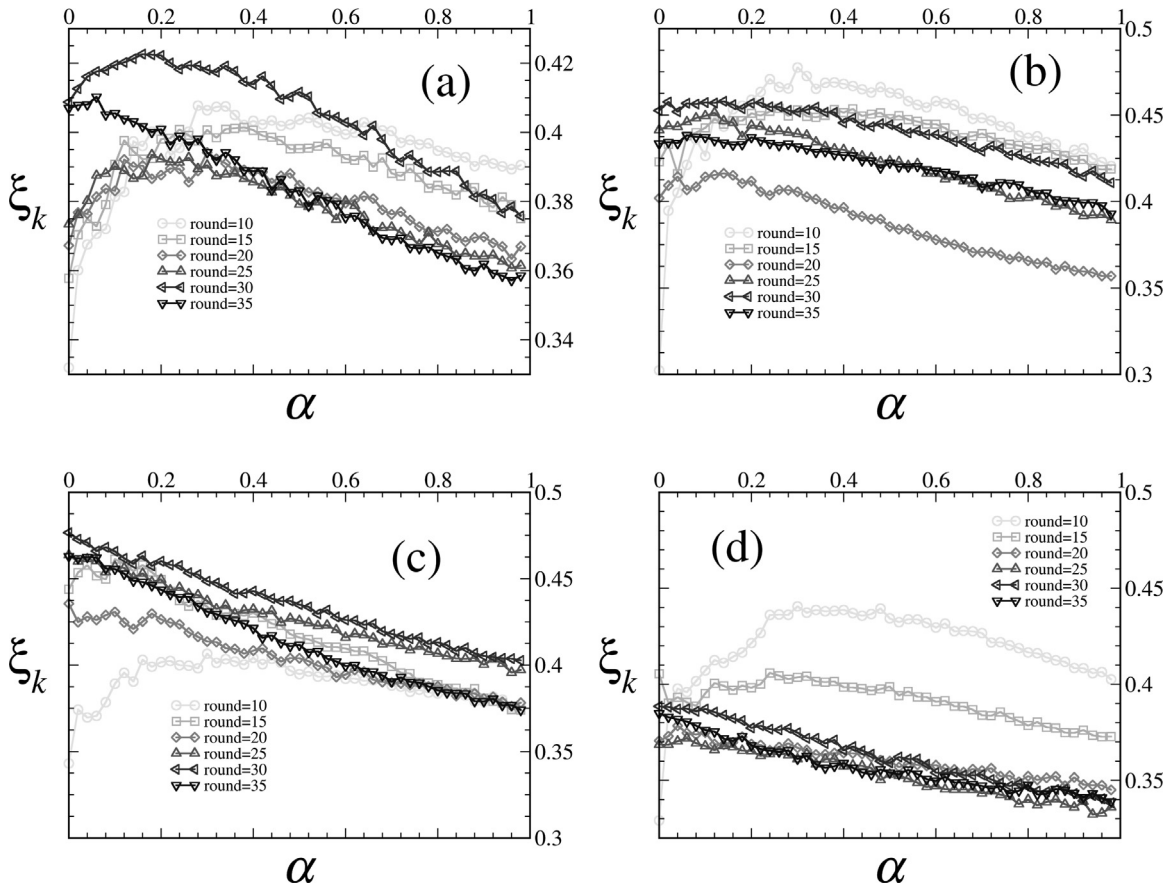


Fig. 3. Matching hits metric versus α for different rounds: $k = 10, 15, 20, 25, 30$, and 35 . Plots (a), (b), (c), and (d), shows the results for the seasons 2020, 2019, 2018, and 2017, respectively. In all situations, we used $\Delta k = 10$, $N_{run} = 1000$, and $\beta = \gamma = 1$.

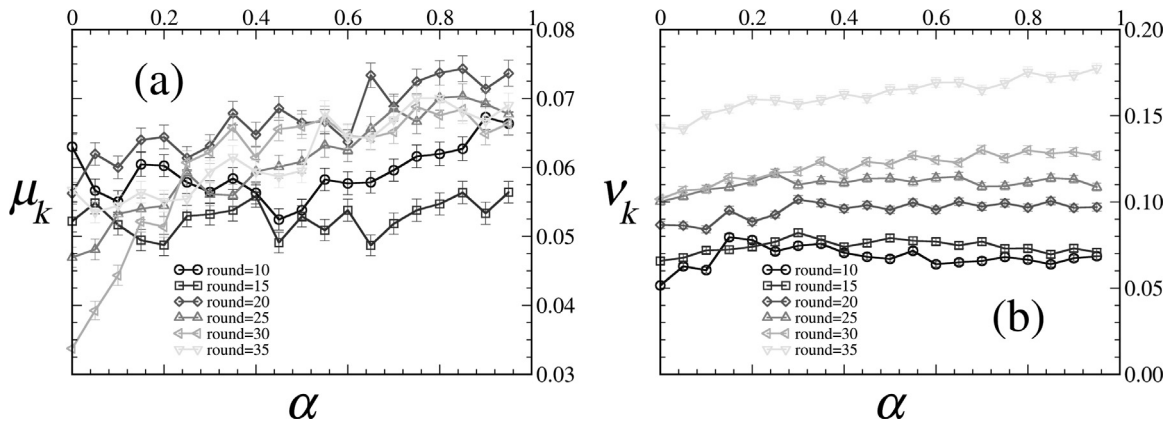


Fig. 4. The score-based (a) and ranking-based (b) quantities μ_k and ν_k , respectively, as function of α for different rounds k of the 2020 *Brasileirão* season.

Once we understand the essential aspects of the optimization part of our model, a question to answer is whether such optimal coefficients can bring good predictions for the championship's final results. As we will show below, the answer is yes and the optimal parameters used as input in Algorithm 3 can deliver good predictions to the final results of the championships.

To study the predictive abilities of our model, we fed Algorithm 3 with data of 2017, 2018, 2019, and 2020 seasons of the main Brazilian football league in order to compare the outcomes of our simulations with the final results of

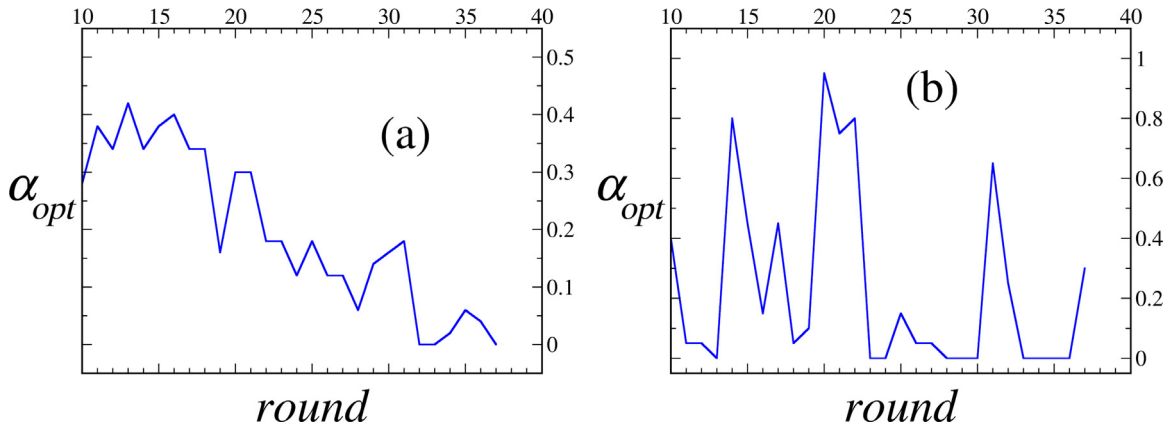


Fig. 5. Evolution of the parameter α_{opt} over rounds while keeping $\beta = \gamma = 1$ for the 2020 season of the main Brazilian football league. (a) corresponds to our estimates obtained when using the matching hits criterion and (b) when considering, as criterion, the maximization of $\mu + \nu$.

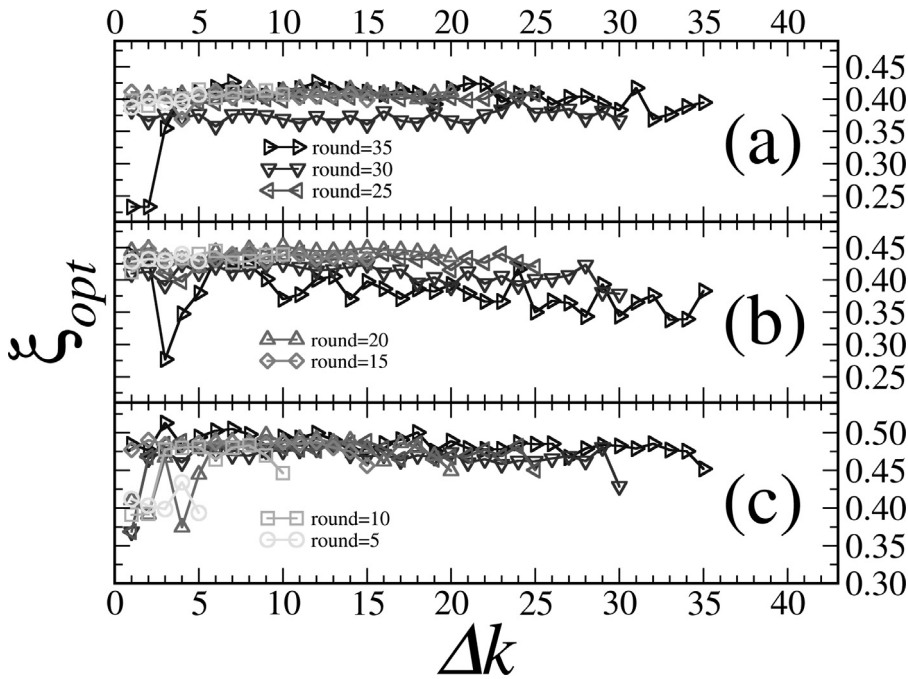


Fig. 6. Optimal ξ_k denoting the average matching hits obtained with α_{opt} as function of Δk for 2018 (a), 2019 (b), and 2020 (c), editions of the Brazilian championship.

each season. We should point out here, once more, that the 2020 season will deserve more attention because it was a particularly complex and decided only in the last round and held during the first year of the Covid-19 pandemic.

As mentioned throughout the paper, we are not only concerned with the champion of the league. The intent of the model is also to be able to predict the top four teams (usually denoted as g4) since their positions in the standings qualify them to the stage groups of the continental cup known as Conmebol Libertadores (Spanish and Portuguese for liberators) cup⁶ in the case of South American clubs. We also made predictions for the last four teams of the standings (usually denoted as z4) once they are relegated to a lower league division (Brazilian Championship B Series) in the next season. For that, we define the following variables:

$$f_w = \frac{1}{N_{run}} \sum_{i=1}^{N_{run}} \theta_i, \quad f_{g4} = \frac{1}{4N_{run}} \sum_{i=1}^{N_{run}} \iota_i, \quad \text{and} \quad f_{z4} = \frac{1}{4N_{run}} \sum_{i=1}^{N_{run}} \kappa_i, \quad (11)$$

⁶ <http://www.conmebol.com/pt-br/torneos/conmebol-libertadores>.

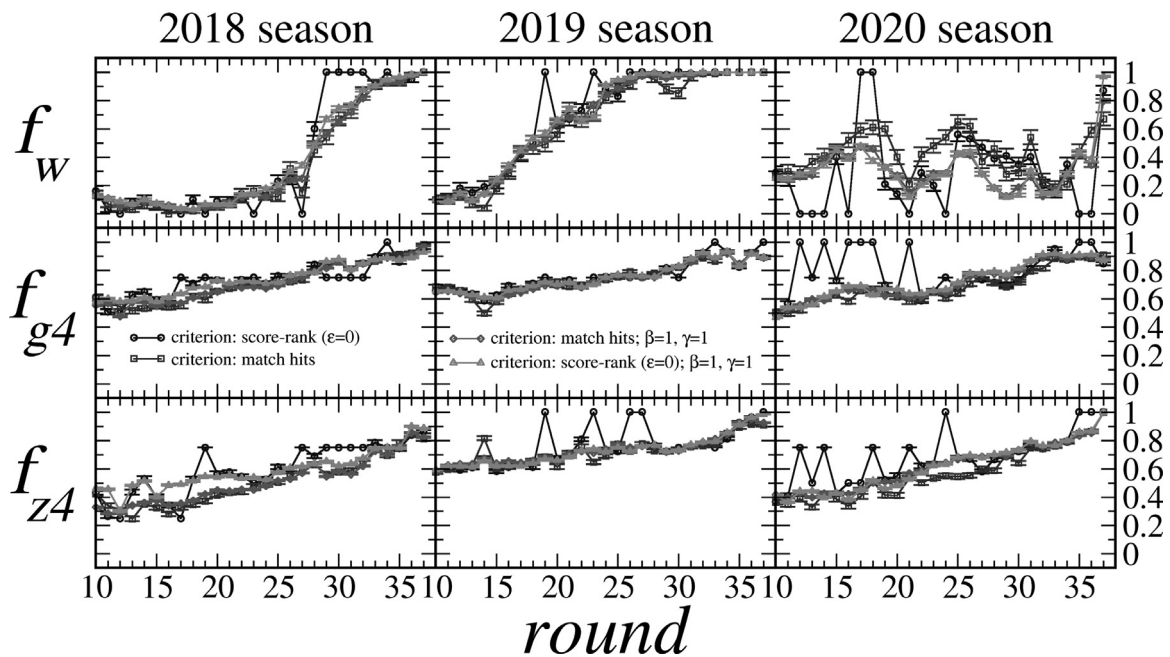


Fig. 7. Evolution of predictive variables f_w , f_{g4} , and f_{z4} for the 2018, 2019, and 2020 seasons of the *Brasileirão*. All situations lead to good predictions with a more oscillating evolution for the score-ranking metric when using the optimization of the three coefficients simultaneously. When considering the matching hits metric or by fixing $\beta = \gamma = 1$, the oscillation is reduced.

Table 1

Hit estimates of the proposed model in relation to the champion of the Brazilian Championship A Series in 2017, 2018, and 2019 seasons compared with the results obtained by SPI. Here, (*) denotes the case where we fixed $\beta = \gamma = 1.0$.

Season	2017			2018			2019		
k	25	30	35	25	30	35	25	30	35
Score-ranking metric	1.00	1.00	1.00	0.23	1.00	0.93	0.83	1.00	1.00
Score-ranking metric (*)	0.87	0.76	1.00	0.20	0.74	0.96	0.94	0.99	1.00
Matching hits metric	0.67	0.54	1.00	0.12	0.67	0.93	0.89	0.85	1.00
Matching hits metric (*)	0.75	0.68	1.00	0.16	0.65	0.92	0.88	0.97	1.00
SPI	0.78	0.74	0.99	0.26	0.77	0.97	0.94	0.97	1.00

where θ is equal to 1 if the simulation got the actual champion and 0 otherwise, ι and κ can assume integer values in the range $[0, 4]$ depending on the number of teams that the simulations successfully hit in the $g4$ and $z4$ regions, respectively, of the standings.

In Fig. 7, we show the evolution of the predictive variables f_w , f_{g4} , and f_{z4} for the 2018, 2019, and 2020 seasons of the Brazilian championship.

The black curve (circle) shows the parameters obtained with score-ranking metric, dark gray (square) corresponds to those obtained with matching hits criterion. Following, gray (diamond) and light-gray (triangle) correspond to matching hits and score-ranking metrics, respectively, when β and γ are set to unity, and only the influence of the market value (α) was optimized in this case. We can observe that score-ranking metric with three optimized parameters presents a more oscillating evolution. All situations predict $g4$, $z4$, and the champion after the simulation of a reasonable number of rounds. As also mentioned above, the prediction of the champion of the 2020 season was more complicated than previous editions and, therefore, will be considered separately in this analysis. To support this statement, we compare our results with Soccer Power Index (SPI) from the project FiveThirtyEight of the *Walt Disney Company* to corroborate this observation. The SPI forecast uses *SPI ratings*, which also takes into account market value and performance statistics in some more complex manner as it can be seen in their webpage.⁷ As we present below, our model could obtain predictions as good or even better than those of SPI with far less complicated steps. Thus, let us summarize them in numbers!

The results obtained for the champion of 2017, 2018, and 2019 seasons are shown in Table 1. For simplicity, we omitted the results for the $g4$ and $z4$ predictions in this table, which means that we have only computed the chances of a given team to win the championship of each specific season.

⁷ <https://projects.fivethirtyeight.com/previsoes-de-futebol/brasileirao/>.

Table 2

Hit estimates of the proposed model in relation to the champion of the Brazilian Championship A Series in 2020 season – during the Covid-19 pandemic. The bold lines stress our best metrics predictions compared with SPI ones. We did not find SPI's predictions for the rounds 29, 31, and 34. Here (*) denotes the case where we fixed $\beta = \gamma = 1.0$.

<i>k</i>	25	26	27	28	29	30	31	32	33	34	35
Score-ranking metric	0.56	0.53	0.47	0.39	0.41	0.35	0.40	0.20	0.15	0.35	0.00
Score-ranking metric (*)	0.42	0.43	0.28	0.18	0.11	0.13	0.30	0.14	0.14	0.28	0.43
Matching hits metric	0.65	0.62	0.37	0.46	0.28	0.29	0.54	0.24	0.19	0.21	0.44
Matching hits metric (*)	0.42	0.41	0.28	0.18	0.12	0.18	0.26	0.12	0.17	0.27	0.42
SPI	0.34	0.43	0.43	0.28	–	0.16	–	0.23	0.15	–	0.25

Table 3

Bundesliga – season 2018–2019: Bayern Munich has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Score-ranking metric	0.59	0.00	0.08	0.14	0.05	0.08	0.15	0.42	0.39	0.44	0.00	0.15	0.00	0.39	0.66	0.89	1.00	0.00	1.00	0.83	0.56	1.00	0.96	0.94
Score-ranking metric (*)	0.52	0.13	0.05	0.06	0.07	0.05	0.20	0.28	0.28	0.20	0.08	0.20	0.48	0.40	0.74	0.73	0.74	0.43	0.77	0.69	0.68	0.89	0.98	0.98
Matching hits metric	0.59	0.42	0.29	0.30	0.21	0.27	0.58	0.50	0.46	0.53	0.47	0.57	0.68	0.67	0.86	0.89	0.82	0.58	0.83	0.76	0.79	0.82	0.99	0.88
Matching hits metric (*)	0.55	0.36	0.22	0.23	0.17	0.19	0.36	0.33	0.29	0.30	0.25	0.35	0.46	0.42	0.68	0.75	0.69	0.51	0.74	0.73	0.68	0.84	0.95	0.90
SPI	0.79	0.70	0.55	0.44	0.45	0.45	–	0.61	0.65	0.64	0.63	0.52	0.64	0.73	0.66	0.84	0.85	0.83	0.69	0.87	0.90	0.87	0.94	0.98

Table 4

Bundesliga – season 2019–2020: Bayern Munich has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Score-ranking metric	0.10	0.00	0.00	0.10	0.01	0.07	0.07	0.20	0.35	0.59	1.00	1.00	0.60	0.59	1.00	1.00	0.82	0.86	0.96	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.16	0.18	0.19	0.19	0.02	0.05	0.10	0.18	0.40	0.53	0.65	0.61	0.61	0.52	0.73	0.83	0.87	0.94	0.99	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.25	0.33	0.43	0.32	0.20	0.23	0.31	0.39	0.45	0.54	0.55	0.64	0.64	0.69	0.77	0.77	0.81	0.79	0.96	0.97	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.21	0.33	0.43	0.27	0.16	0.20	0.26	0.36	0.41	0.52	0.62	0.63	0.65	0.62	0.72	0.76	0.78	0.82	0.97	0.99	1.00	1.00	1.00	1.00
SPI	0.76	0.60	0.69	0.82	0.71	0.57	–	0.59	0.59	0.64	0.76	0.81	0.78	0.77	0.73	0.85	0.88	0.92	–	0.99	0.99	0.99	1.00	1.00

Table 5

Bundesliga – season 2020–2021: Bayern Munich has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Score-ranking metric	0.62	0.44	0.00	1.00	0.69	0.48	1.00	0.84	1.00	1.00	1.00	0.96	0.55	1.00	0.69	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.54	0.32	0.30	0.59	0.67	0.59	0.82	0.89	0.97	0.97	0.97	0.94	0.75	0.70	0.81	0.93	0.97	1.00	0.98	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.64	0.26	0.56	0.68	0.70	0.45	0.58	0.57	0.88	0.94	0.93	0.91	0.81	0.60	0.84	0.87	0.88	0.98	0.93	1.00	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.57	0.34	0.47	0.62	0.64	0.62	0.47	0.74	0.86	0.89	0.73	0.79	0.66	0.66	0.69	0.70	0.71	0.93	0.87	0.97	1.00	1.00	1.00	1.00
SPI	0.79	0.76	–	0.68	0.82	0.82	0.79	–	0.82	0.93	0.92	0.93	0.88	0.81	0.78	0.84	0.91	0.89	0.98	0.95	–	>0.99	>0.99	1.00

Our results also present strong correlations with those of SPI (displayed in the last line of Table 1), showing that our predictions are as good as the SPI predictions. The similarities of both predictions are not exactly a surprise. However, when we remember that we are using a performance index (scored points) worst than scored goals [33] or chances of goals [34] to make predictions, and far less detailed than the *SPI rating*, we realize how robust our algorithm is.

Finally, we would like to look into the numbers of a *sui generis* season: 2020. This season was held during the first year of the Covid-19 pandemic, which led to the absence of fans in the stadium and the champion (Flamengo⁸) was known only at the end of the last round,⁹ affecting dramatically the predictions. To get an idea, Flamengo conquered the leadership in the penultimate round. However in the last round, Flamengo (away team) lost its match against São Paulo¹⁰ (home team), but Internacional¹¹ (home team), which was leading the championship since the 31st round, only drew against Corinthians¹² (away team). The aftermath: Flamengo won the championship with just 71 scored points, one point ahead of the runner-up, Internacional, which needed a straightforward victory against Corinthians in the last round to become champion, making this, one of the most exciting seasons of the Brazilian football league in recent years.

As we can observe in Table 2, the matching hits metric with optimization of three coefficients has better predictive performance than the SPI predictions for the entire 2020 season of the *Brasileirão*, except for round 27. The score-rating metric results show strong fluctuations (we can see them at the f_w vs *round* plot of the 2020 season in Fig. 7) and relative worst predictive performance than SPI forecasts. When using the matching hits metric with only one optimized coefficient, our results show a slightly worst predictive performance than SPI's. However, it presents fewer fluctuations in comparison with the other metric. Based on our predictions using three optimized coefficients with the matching hits metric, we conclude that our results are remarkable considering our model's simplicity.

⁸ <https://www.flamengo.com.br/>.

⁹ <https://www.cbf.com.br/futebol-brasileiro/competicoes/campeonato-brasileiro-serie-a/2020>.

¹⁰ <http://www.saopaulofc.net/>.

¹¹ <https://internacional.com.br/>.

¹² <https://www.corinthians.com.br/>.

Table 6

Calcio-Serie A – season 2018–2019: Juventus has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.62	1.00	0.72	0.76	0.79	0.89	0.92	0.92	0.97	1.00	0.98	1.00	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.49	0.51	0.64	0.76	0.78	0.86	0.88	0.89	0.90	0.97	0.96	0.99	0.98	0.95	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.56	0.69	0.81	0.84	0.84	0.90	0.95	0.93	0.99	0.96	0.92	0.97	0.96	1.00	0.98	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.54	0.55	0.66	0.75	0.82	0.90	0.89	0.92	0.91	0.93	0.90	0.95	0.89	0.94	0.97	0.97	0.99	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SPI	0.73	0.68	0.77	0.81	0.83	0.86	0.88	0.92	0.96	0.87	0.85	0.86	0.84	0.79	0.89	0.92	0.96	0.97	>0.99	–	0.98	>0.99	>0.99	–	>0.99	1.00	1.00	1.00

Table 7

Calcio-Serie A – season 2019–2020: Juventus has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.52	0.70	1.00	0.75	0.55	0.29	0.56	0.47	0.50	1.00	0.54	1.00	1.00	1.00	1.00	0.55	0.60	1.00	0.84	0.75	0.93	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.53	0.53	0.62	0.60	0.41	0.32	0.46	0.49	0.53	0.59	0.83	0.69	0.76	0.48	0.56	0.56	0.67	0.83	0.80	0.87	0.93	1.00	0.99	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.56	0.51	0.55	0.78	0.59	0.59	0.66	0.62	0.51	0.71	0.85	0.69	0.81	0.65	0.83	0.80	0.89	0.91	0.83	0.90	0.97	0.98	1.00	0.95	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.51	0.52	0.60	0.60	0.48	0.44	0.47	0.51	0.50	0.61	0.69	0.67	0.69	0.52	0.45	0.55	0.67	0.72	0.71	0.81	0.91	0.95	0.97	0.96	1.00	1.00	1.00	1.00
SPI	0.21	0.27	0.34	0.41	0.43	0.41	0.41	0.44	0.38	0.49	0.52	0.64	0.70	0.72	0.58	0.40	0.58	0.38	–	0.32	–	0.46	–	0.77	–	0.84	0.95	1.00

Table 8

Calcio-Serie A – season 2020–2021: Internazionale Milano has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.00	0.22	0.00	0.46	0.30	0.56	0.36	0.35	0.00	0.00	0.00	0.00	0.64	0.83	0.80	0.86	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.18	0.20	0.38	0.40	0.42	0.38	0.30	0.24	0.33	0.30	0.26	0.35	0.69	0.89	0.77	0.83	0.91	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.19	0.25	0.40	0.41	0.60	0.52	0.40	0.29	0.40	0.39	0.34	0.42	0.59	0.76	0.81	0.83	0.95	0.92	0.94	0.98	0.98	1.00	0.99	1.00	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.17	0.21	0.36	0.38	0.41	0.37	0.33	0.22	0.30	0.30	0.28	0.35	0.55	0.70	0.75	0.77	0.84	0.90	0.84	0.99	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1.00
SPI	0.25	0.32	0.39	0.29	–	0.37	0.42	0.44	0.48	0.48	0.51	0.60	0.49	0.52	0.40	0.42	0.42	0.35	0.28	0.18	0.18	0.29	–	0.16	0.31	–	0.80	0.75

Table 9

La Liga – season 2018–2019: Barcelona FC has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.52	0.62	1.00	0.42	0.63	0.72	0.47	0.73	1.00	0.91	0.88	0.92	0.86	1.00	0.90	0.91	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.55	0.63	0.42	0.48	0.61	0.66	0.62	0.76	0.85	0.86	0.90	0.85	0.88	0.77	0.89	0.94	0.96	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.65	0.57	0.32	0.57	0.75	0.76	0.75	0.87	0.83	0.94	0.95	0.93	0.93	0.92	0.97	0.95	0.96	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.62	0.66	0.47	0.50	0.65	0.71	0.74	0.80	0.89	0.86	0.83	0.85	0.83	0.73	0.82	0.88	0.94	0.93	0.98	0.97	0.97	1.00	0.99	0.99	1.00	1.00	1.00	1.00
SPI	0.54	0.64	0.67	0.53	0.60	0.61	0.69	0.74	0.75	0.86	0.87	0.85	0.86	0.84	0.79	0.89	0.92	0.96	0.97	>0.99	–	0.98	>0.99	>0.99	–	>0.99	1.00	1.00

Table 10

La Liga – season 2019–2020: Real Madrid Club de Fútbol has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.24	0.34	0.27	0.28	0.27	0.00	0.38	0.37	0.19	0.00	0.55	1.00	0.68	0.75	0.54	0.00	0.66	0.00	0.00	0.00	0.00	0.50	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.19	0.29	0.22	0.15	0.18	0.30	0.39	0.41	0.29	0.42	0.49	0.71	0.71	0.80	0.54	0.32	0.68	0.27	0.26	0.33	0.57	0.40	0.72	0.88	0.93	0.88	0.95	1.00
Matching hits metric	0.32	0.35	0.33	0.28	0.36	0.33	0.43	0.40	0.26	0.40	0.50	0.71	0.77	0.70	0.56	0.55	0.59	0.48	0.38	0.40	0.48	0.53	0.54	0.84	0.97	0.89	1.00	1.00
Matching hits metric (*)	0.32	0.35	0.31	0.37	0.38	0.37	0.39	0.38	0.31	0.42	0.54	0.64	0.69	0.70	0.53	0.51	0.62	0.49	0.34	0.41	0.54	0.55	0.66	0.83	0.87	0.87	0.92	1.00
SPI	0.21	0.27	0.34	0.41	0.43	0.41	0.41	0.44	0.38	0.49	0.52	0.64	0.70	0.72	0.58	0.40	0.58	0.38	–	0.32	–	0.46	–	0.77	–	0.84	0.95	1.00

Table 11

La Liga – season 2020–2021: Club Atlético de Madrid has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.49	0.51	1.00	0.44	0.41	1.00	0.75	0.71	0.88	0.92	0.96	0.96	0.98	0.89	0.70	1.00	1.00	1.00	0.92	0.57	1.00	1.00	1.00	1.00	1.00	1.00	0.88	0.90
Score-ranking metric (*)	0.49	0.60	0.73	0.46	0.44	0.51	0.76	0.78	0.86	0.91	0.95	0.97	0.94	0.88	0.80	0.86	0.89	0.87	0.79	0.68	0.78	0.91	0.64	0.74	0.55	0.65	0.90	0.95
Matching hits metric	0.58	0.61	0.68	0.49	0.50	0.61	0.77	0.76	0.91	0.90	0.88	0.95	0.96	0.82	0.76	0.85	0.90	0.82	0.81	0.60	0.44	0.49	0.60	0.59	0.56	0.49	0.97	1.00
Matching hits metric (*)	0.49	0.66	0.67	0.53	0.58	0.63	0.73	0.72	0.88	0.88	0.88	0.93	0.89	0.85	0.74	0.79	0.81	0.79	0.72	0.52	0.62	0.70	0.63	0.54	0.55	0.57	0.87	0.91
SPI	0.25	0.32	0.39	0.29	–	0.37	0.42	0.44	–	0.48	0.51	0.60	0.49	0.52	0.40	0.42	0.42	0.35	0.28	0.18	0.18	0.29	0.16	–	0.31	–	0.80	0.75

Table 12

Premier League – season 2018–2019: Manchester City FC has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.25	0.43	0.40	0.37	0.38	0.53	0.00	0.37	0.00	0.19	0.13	0.00	0.17	0.14	0.14	0.22	0.39	0.44	0.37	1.00	1.00	0.41	0.41	0.18	0.25	0.73	0.72	0.78
Score-ranking metric (*)	0.27	0.31	0.32	0.47	0.52	0.53	0.34	0.37	0.22	0.10	0.09	0.15	0.15	0.16	0.10	0.16	0.37	0.40	0.41	0.65	0.68	0.40	0.57	0.26	0.18	0.65	0.77	0.79
Matching hits metric	0.26	0.39	0.41	0.43	0.55	0.43	0.39	0.33	0.20	0.15	0.11	0.23	0.19	0.23	0.08	0.23	0.41	0.50	0.38	0.49	0.56	0.52	0.54	0.31	0.20	0.64	0.60	0.84
Matching hits metric (*)	0.29	0.32	0.45	0.44	0.44	0.49	0.36	0.40	0.20	0.10	0.12	0.21	0.20	0.18	0.11	0.18	0.27	0.33	0.34	0.52	0.59	0.48	0.48	0.36	0.34	0.53	0.73	0.74
SPI	0.64	0.70	0.74	0.73	–	0.74	0.60	0.53	–	0.20	–	0.25	0.24	0.29	0.24	0.40	0.49	0.47	0.49	0.65	0.67	0.66	0.60	0.56	0.52	0.73	0.82	0.88

3.2. Validation of the algorithm: European leagues

Finally, we will conclude our results by presenting our estimates for four European championships of international importance: Calcio in Italy, Bundesliga in Germany, Premier League in England, and La Liga in Spain, for the seasons 2018–2019, 2019–2020, 2020–2021, in order to validate the method showing its robustness in predicting the final results of championships.

To reach this goal, we applied our method for these championships as we have done for the Brazilian championship. Since the method was extensively employed to the Brazilian league, we only present the summarized results for the

Table 13

Premier League – season 2019–2020: Liverpool FC has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.75	1.00	1.00	1.00	0.97	0.99	0.98	0.99	1.00	1.00	0.98	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Score-ranking metric (*)	0.74	0.87	0.95	0.95	0.97	0.96	0.99	1.00	1.00	0.99	0.98	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric	0.82	0.76	0.81	0.81	0.94	0.89	0.99	0.94	0.95	1.00	0.98	1.00	1.00	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Matching hits metric (*)	0.77	0.75	0.84	0.88	0.89	0.91	0.93	0.91	0.93	0.96	0.99	0.95	0.98	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SPI	0.53	0.52	0.67	0.69	-	0.70	0.85	0.84	0.82	-	-	0.97	0.98	-	>0.99	>0.99	>0.99	>0.99	>0.99	>0.99	>0.99	>0.99	>0.99	1.00	1.00	1.00	1.00	1.00

Table 14

Premier League – season 2020–2021: Manchester City FC has won the championship.

<i>k</i>	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
Score-ranking metric	0.09	0.13	0.12	0.10	0.05	0.10	0.13	0.13	0.18	0.00	0.35	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	
Score-ranking metric (*)	0.11	0.11	0.13	0.08	0.08	0.17	0.12	0.12	0.12	0.30	0.41	0.71	0.82	0.95	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	
Matching hits metric	0.14	0.15	0.20	0.08	0.06	0.19	0.17	0.29	0.34	0.54	0.57	0.61	0.71	0.85	0.97	0.97	0.99	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	
Matching hits metric (*)	0.15	0.12	0.13	0.08	0.09	0.18	0.15	0.23	0.26	0.41	0.45	0.48	0.60	0.75	0.93	0.95	0.99	0.99	1.00	0.99	1.00	0.99	0.99	1.00	1.00	1.00	1.00	
SPI	0.51	0.51	–	0.46	0.42	–	0.54	0.69	0.70	0.76	0.87	–	0.88	0.95	0.99	>0.99	>0.99	>0.99	–	>0.99	>0.99	>0.99	>0.99	>0.99	>0.99	1.00	1.00	1.00

European championships since the plots and preparatory developments follow the same picture as shown above in this section.

The obtained results were so good as the Brazilian championship as can be observed in [Tables 3–5](#) of Bundesliga. In [Tables 6–8](#), we present the results for the Italian league and in [Tables 9–11](#), La Liga championship results are showed. Finally, [Tables 12–14](#) present the results obtained for the Premier league.

In all of these tables, we also present the results obtained by SPI predictions and, as can be notice, our estimates are befitting with the SPI ones.

4. Summaries, conclusions, and discussions

Our algorithm was capable of determining essential statistics of the real Brazilian Championship A Series and of four European leagues (La Liga, Calcio, Premier, and Bundesliga). We initially presented the results of Brazilian league with suitable graphical analysis showing how the method works. Following, we only reproduce the results using the same techniques to obtain the final statistics for the European leagues since the investigation is entirely similar.

Our predictions are compatible with an alternative and commercial way to perform predictions (SPI) since, as shown, both predictions are strongly correlated to each other. On the other hand, our method presented more reliable estimates in a more complicated season: 2020. Our additional idea is to take a different path from SPI, making explicit all of our parameters. The model uses many concepts related to the teams' scoring process during a championship and the market values of the teams.

Estimates for g_4 and z_4 show notable agreement between the metrics considered when the predictions are performed from the 26th round on. In fact, from this round on, one obtains something between 80%–100% of probability to point out the exact g_4 or z_4 (see [Fig. 7](#)). We believe that we can change our predictive method to contemplate other championships not based on the double-round robin system scheme. Other authors [\[36\]](#) explored machine learning for the Premier League in 2014 and 2015 seasons with good predictions and the topic seems to be extremely promising for future works by including other sports.

CRediT authorship contribution statement

Eduardo Velasco Stock: Conceived and designed the analysis, Data curation, Formal analysis, Wrote the paper, Elaborated the algorithms. **Roberto da Silva:** Conceived and designed the analysis, Data curation, Formal analysis, Wrote the paper, Elaborated the algorithms. **Henrique A. Fernandes:** Conceived and designed the analysis, Wrote the paper, Revised the results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] R. Scarfe, C. Singleton, P. Telemo, *Ind. Relat.* 60 (2021) 84–118.
- [2] C. Lucifora, R. Simmons, *J. Sports Econ.* 4 (1) (2003) 35–55.
- [3] F. Alvaredo, A.B. Atkinson, T. Piketty, E. Saez, *J. Econ. Perspect.* 27 (3) (2013) 3–20.
- [4] L. Uehara, M. Falcous, C. Button, K. Davids, D. Araújo, A.R. de Paula, J. Saunders, *Front. Sports Active Living* 3 (2021).
- [5] A. Heuer, O. Rubner, *Eur. Phys. J. B* 67 (2009) 445.
- [6] G. Skinner, G. Freeman, *J. Appl. Stat.* 36 (2009) 1087.
- [7] H. Ribeiro, R. Mendes, L. Malacarne, S. Piccoli Jr., P. Santoro, *Eur. Phys. J. B* 75 (2010) 327.
- [8] J. Lasek, M. Gagolewski, *Stat. Model.* 18 (5–6) (2018) 411–435.
- [9] E. Ben-Naim, N.W. Hengartner, *Phys. Rev. E* 76 (2007) 026106.
- [10] E. Ben-Naim, N.W. Hengartner, S. Redner, F. Vazquez, *J. Stat. Phys.* 151 (2013) 458–474.
- [11] Igor. Zolkernevic, Why upset victories are common in the Brazilian soccer championship, *Pesquisa Fapesp* 212 (2013).
- [12] R. da Silva, M.H. Vainstein, L.C. Lamb, S.D. Prado, *Comput. Phys. Comm.* 184 (2013) 661.
- [13] D. Matesanz, F. Holzmayer, B. Torgler, S.L. Schmidt, G.J. Ortega, *PLoS ONE* 13 (12) (2018) e0209362.
- [14] B.L. Boulier, H.O. Stekler, *Int. J. Forecast.* 19 (2) (2003) 257–270.
- [15] R.P. Bunker, F. Thabtah, *Appl. Comput. Inform.* 15 (1) (2019) 27–33.
- [16] Yu-Chia Hsu, *Appl. Sci.* 11 (14) (2021) 6594.
- [17] R. da Silva, M.H. Vainstein, S. Gonçalves, F.S.F. Paula, *Phys. Rev. E* 88 (2013) 022136.
- [18] R. da Silva, S.R. Dahmen, *Physica A* 398 (2014) 56.
- [19] R. Albert, A.-L. Barabási, *Rev. Modern Phys.* 74 (2002) 47.
- [20] R. da Silva, *Physica A* 500 (2018) 60–71.
- [21] M.J. Maher, *Stat. Neerl.* 36 (1982) 3.

- [22] A. Heuer, C. Muller, O. Rubner, *Eur. Phys. Lett.* 89 (2010) 38007.
- [23] A. Gabel, S. Redner, *J. Quant. Anal. Sports* 8 (1) (2012).
- [24] A.C. Thomas, *J. Quant. Anal. Sports* 3 (3) (2007).
- [25] R. Pollard, *Open Sports Sci. J.* 1 (2008) 12–14.
- [26] R. Pollard, *J. Sports Sci.* 4 (1986) 237–248.
- [27] C. Goumas, *J. Sport Health Sci.* 6 (2017) 321.
- [28] H. Oberhofer, T. Philippovich, H. Winner, *J. Econ. Psychol.* 31 (2010) 200–211.
- [29] F. Wunderlich, M. Weigelt, R. Rein, D. Memmert, *PLoS ONE* 16 (3) (2021) e0248590.
- [30] C. Goumas, *Eur. J. Sport Sci.* 14 (sup1) (2014) S243–S249.
- [31] G.P. Nassis, *J. Strength Cond. Res.* 27 (3) (2013) 703–707.
- [32] S. Merritt, A. Clauset, *Sci. Rep.* 3 (2013) 3067.
- [33] A. Heuer, O. Rubner, *Eur. Phys. J. B* 67 (2009) 445–458.
- [34] A. Heuer, O. Rubner, *PLoS ONE* 9 (9) (2014) e104647.
- [35] J.M. Fernandez-Crehuet, J. Rosales-Salas, P. Navarro, *Sport Soc.* 24 (2) (2021) 131–146.
- [36] R. Baboota, H. Kaur, *Int. J. Forecast.* 35 (2019) 741.